ATTENTION-AUGMENTED CNN ARCHITECTURES FOR MULTIPLE SCLEROSIS MRI CLASSIFICATION: A COMPARATIVE STUDY OF ATTENTION MECHANISMS, METRIC LEARNING, AND VISION TRANSFORMERS

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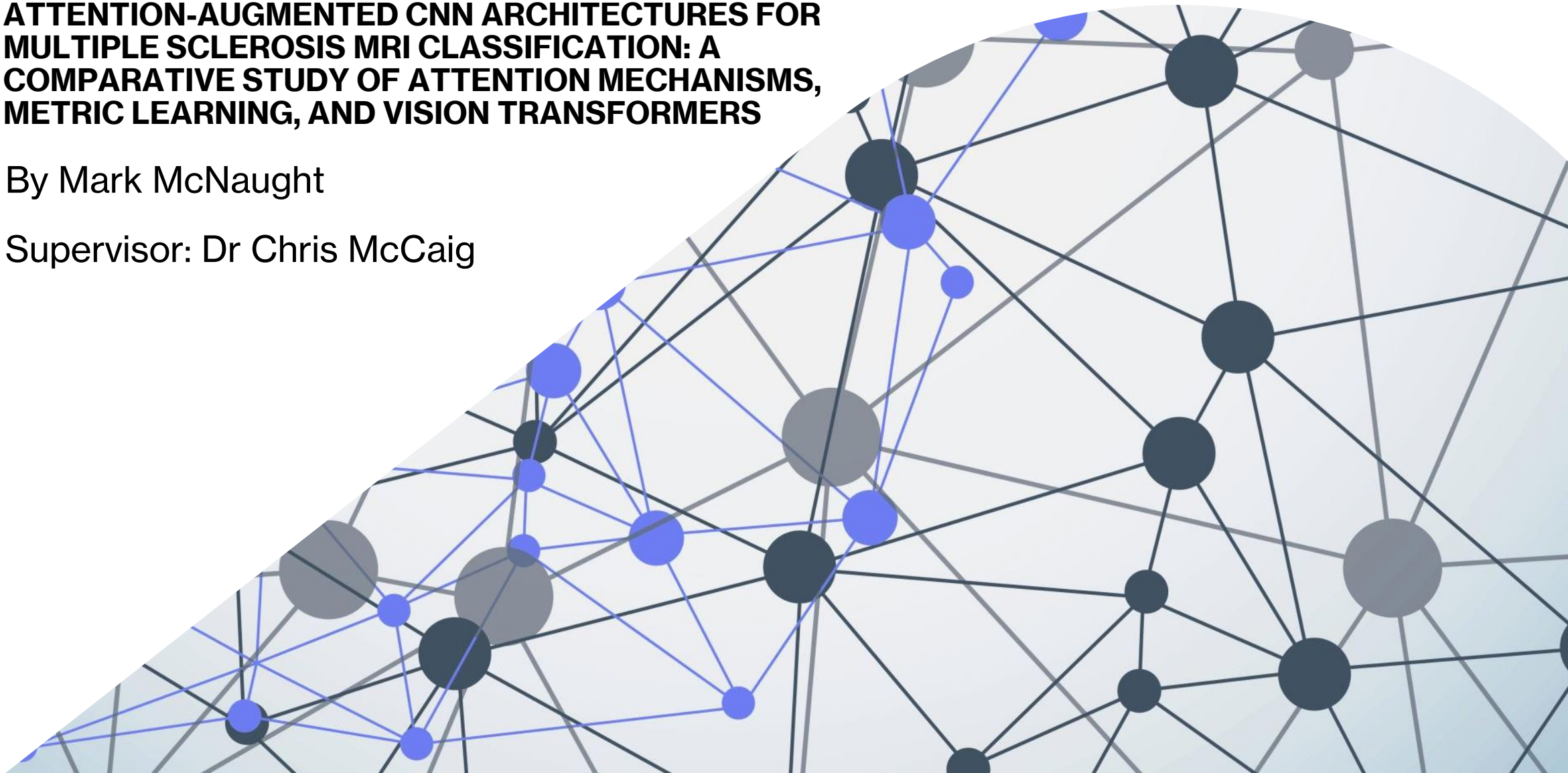


Project Aims

Aim: Systematically evaluate how attention mechanisms, metric learning, and transformer-inspired components extend a pretrained CNN baseline for binary MS classification (under limited-data and compute)

SRQ1 – Attention mechanisms: Do SE-Net channel, CBAM dual, and spatial-isolated attention each improve classification and calibration? And can channel and spatial contributions be separated?

SRQ2 – Metric learning: how does replacing the standard classification head used in SRQ1 with a post-hoc NCA + kNN pipeline impact performance?

SRQ3 – Tokenised Self-Attention: Does a single MHSA layer over ViT-style CNN spatial tokens improve on the baseline classification head?

SRQ4 – Transformer comparison: how does the best CNN variant compare against a parameter-matched Vision Transformer trained under the same protocol?

Multiple Sclerosis (MS)

- Chronic, inflammatory demyelinating disease of the central nervous system
- A leading causes of long-term neurological disability in young adults
- Affects 2.9 million people globally, 150,000 in the UK
- Mean age of onset: ~30 years
- Profound impact on quality of life

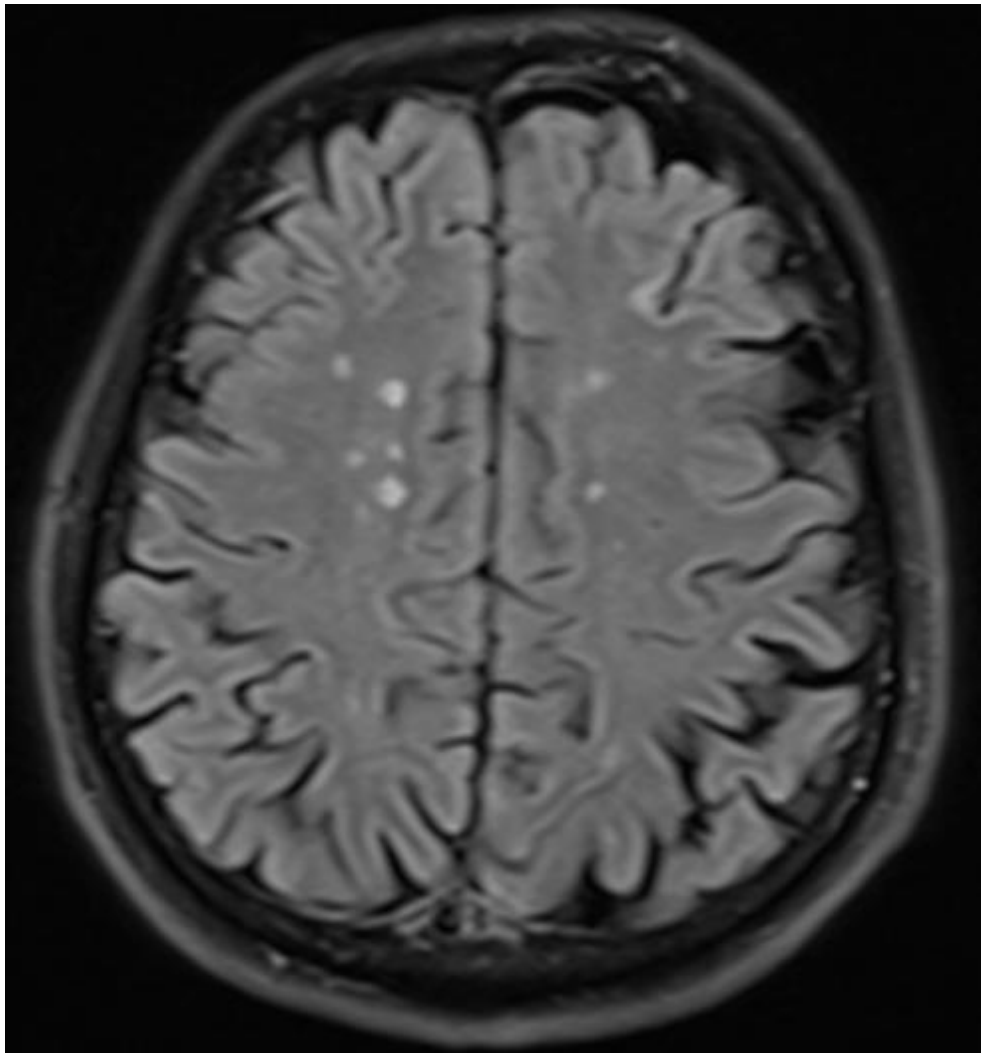


Image 1: MRI axial scan of MS positive patient (Kaggle 2022)

Diagnostic Challenges

- No single definitive diagnostic test (McDonald Criteria), depends on
  - expert interpretation of magnetic resonance imaging (MRI),
    - DiS
    - DiT
  - integrated with clinical history (clinical attacks)
- Misdiagnosed in 15% of cases, underdiagnosed in 36%
- Average diagnostic delay: 17+ months
- Radiologist interpretation is expert-dependant and variable
  - Inter-rater differences of up to 68% in lesion volume estimations
- Traditional methods fail to provide consistent results and generalisation across scans
- Solution: Deep learning - learn representations directly from data
  - Data scarcity issue for small medical datasets
  - Transfer learning

Convolutional Neural Networks (CNNs)

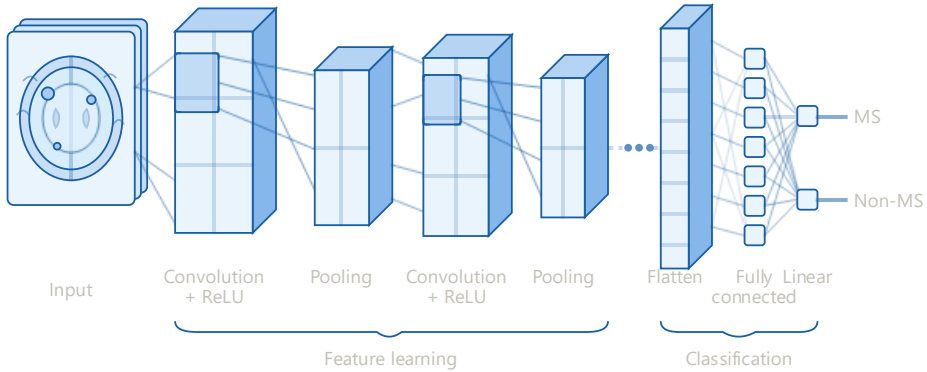


Figure 1: diagram of basic block of a CNN

- Dominant paradigm: 76% of deep learning MS MRI approaches use CNNs
- Spatially structured lesion patterns are a natural fit
- Learnable filters detect local patterns: edges, textures, structures
- Hierarchical representations: low-level features → semantically meaningful features
- Weight sharing: detect a feature once, apply everywhere

Residual Networks (ResNets)

- Adding more layers should improve performance – in practice it makes it worse, even on the data the model was trained on
- ResNets fix this by letting the input skip over each block and rejoin the output – so layers only learn small adjustments rather than everything from scratch
- This keeps gradients flowing back through the network and unlocks training at much greater depths

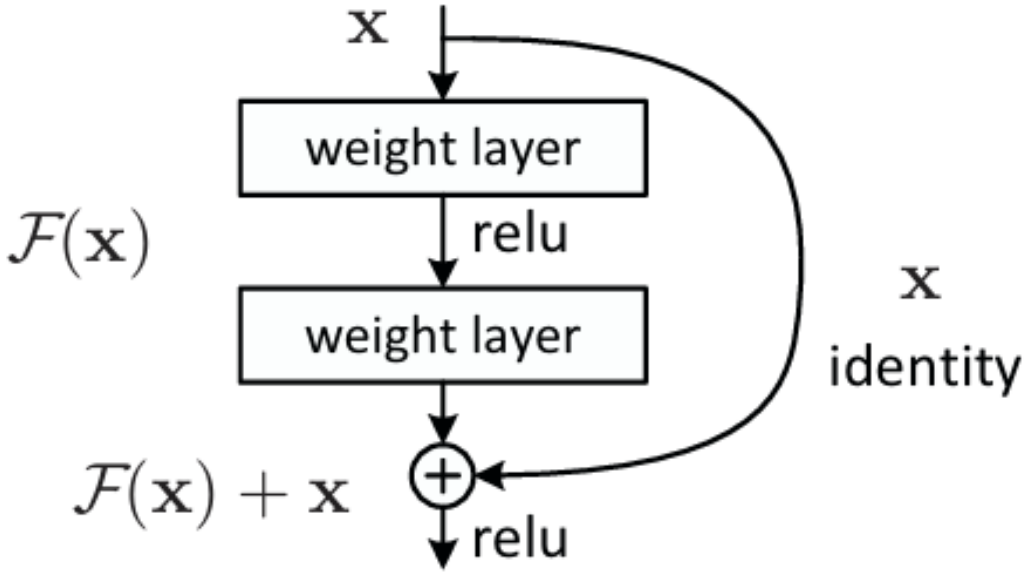


Figure 2: diagram of ResNet basic block (He et al. 2015)

Attention mechanisms

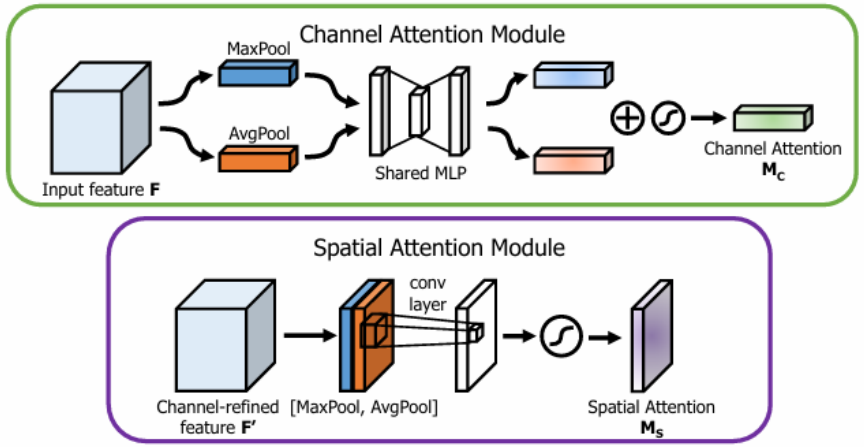


Figure 3: diagram of channel and spatial attention modules (Woo et al. 2018)

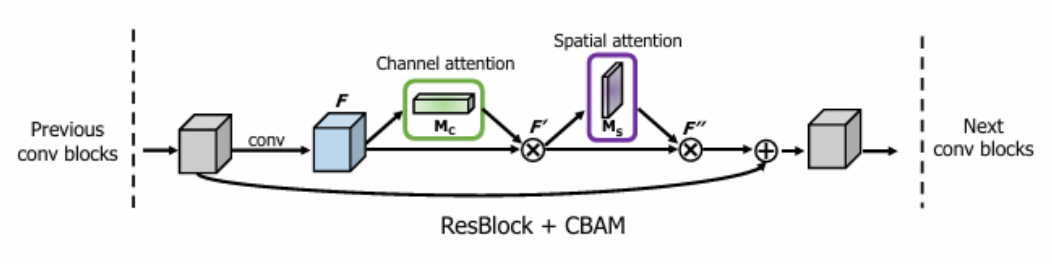


Figure 4: diagram CBAM module (Woo et al. 2018)

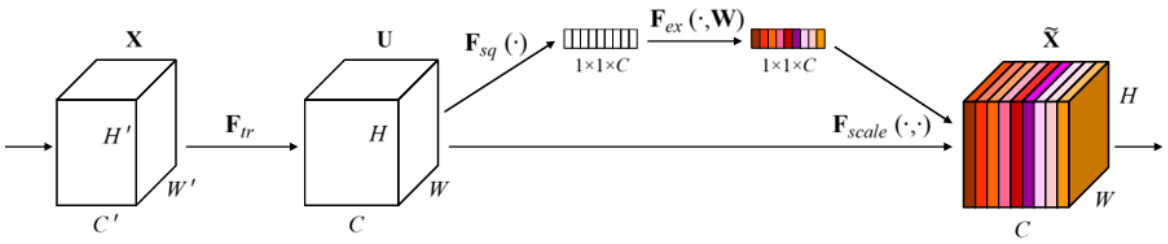


Figure 5: diagram Squee-Excitation block (Hu et al. 2019)

- **Channel:** *what* is important (weights channels)
- **Spatial**: *where* is important (weights regions)
- **End placement**: inserted before the final average pooling layer, operates on the complete feature map once
- **Block-level pre-shortcut**: inserted inside every residual block, after the convolutions but before the skip connection rejoins — gates only what the convolutions learned
- **Block-level post-shortcut**: same position but after the skip connection has been added back, gates the fully merged signal

Metric Learning

- Standard classifiers learn to separate classes – but don't explicitly structure the embedding space by similarity
- Metric learning restructures that space so similar samples cluster together and dissimilar ones are pushed apart
- NCA learns a linear projection of the embedding space that maximises kNN classification accuracy
- Applied post-hoc to frozen backbone embeddings – no retraining required

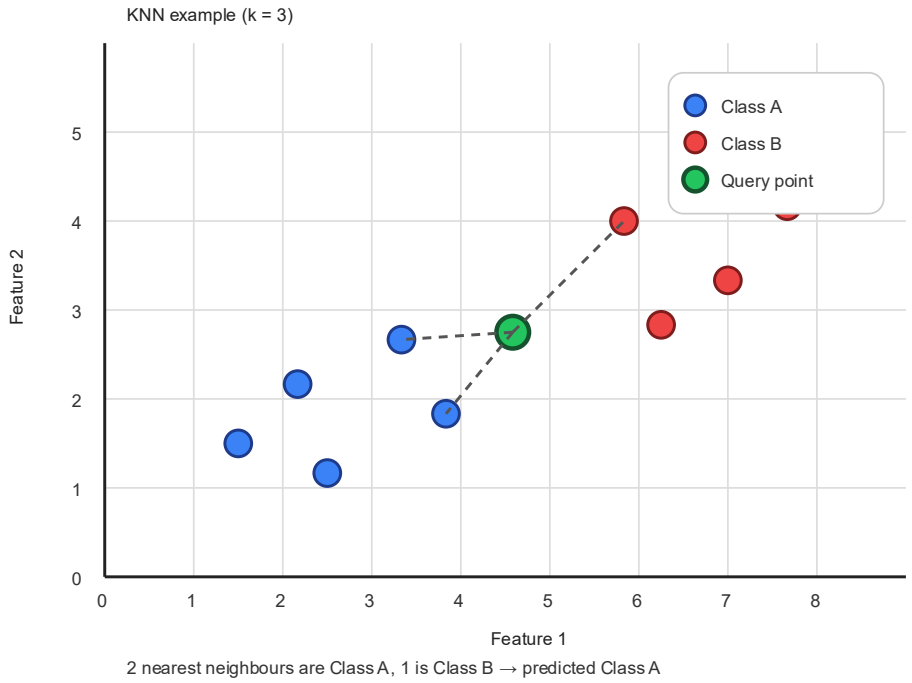


Figure 6: diagram of kNN classification with k=3

Vision Transformers

- Transformers process sequences – ViTs adapt this to images by splitting them into fixed patches, embedding each as a token, and processing the sequence with self-attention
- Self-attention computes relationships between all tokens simultaneously – captures long-range dependencies that CNNs cannot
- Cost: ViTs are data-hungry – without the inductive biases of convolutions, they need far more examples to learn effective representations

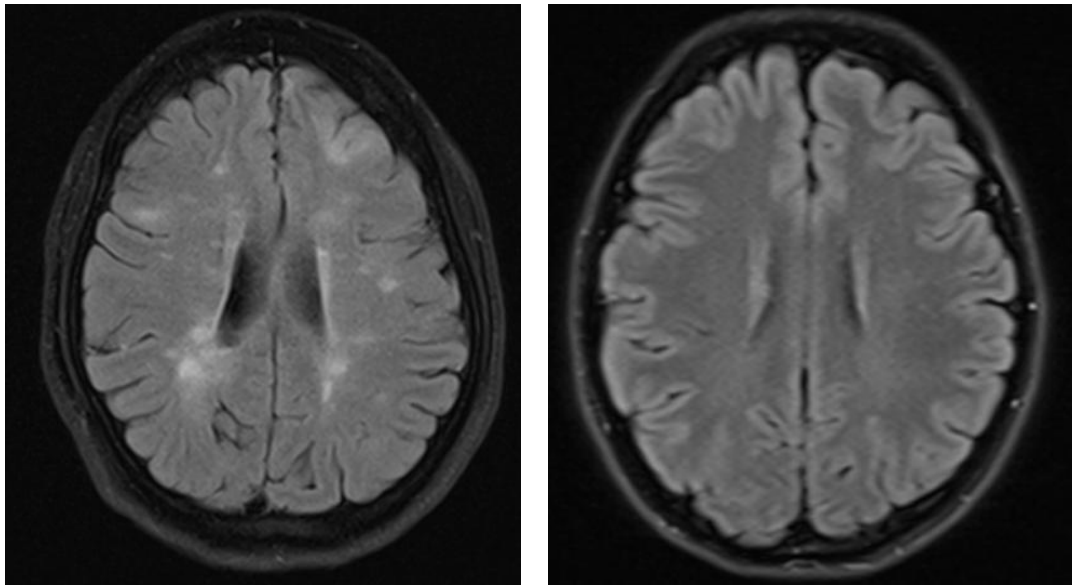


Image 2: MRI axial scans of MS positive and control patients, respectively (Kaggle 2022)

Dataset & Setup

- 1,652 axial FLAIR MRI slices from 131 patients — MS: 650, Control: 1,002
- Stratified 80/20 split → 1,321 train+val, 331 held-out test set aside at the outset
- 5-fold cross-validation for model selection; all experiments CPU-only (Dell XPS 13, 16GB RAM)
- Conservative augmentation: horizontal flip + ±10° rotation only — preserves lesion spatial structure

Performance Metrics

- F1 score – primary metric; balances precision and recall, robust to class imbalance
- AUC-ROC – measures separability across all classification thresholds
- ECE (Expected Calibration Error) – measures how well predicted probabilities reflect true likelihood; lower is better
- Model fold selection: best mean validation F1 across folds
- Final evaluation: single run on held-out test set, used once after all model selection decisions made

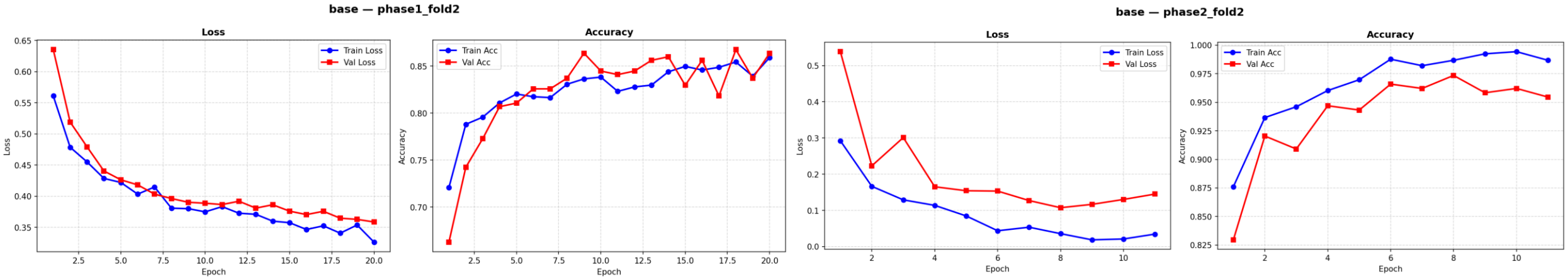


Figure 7: ResNet18 base training curves for fold 2/5

Two-Phase Transfer Learning

- Phase 1: backbone frozen — only classifier head and attention modules trained Phase 2: all parameters unfrozen — full fine-tuning at reduced LR with weight decay for regularisation
- Prevents randomly initialised attention modules from corrupting pretrained features before they've learned anything useful

Experimental Design Decisions

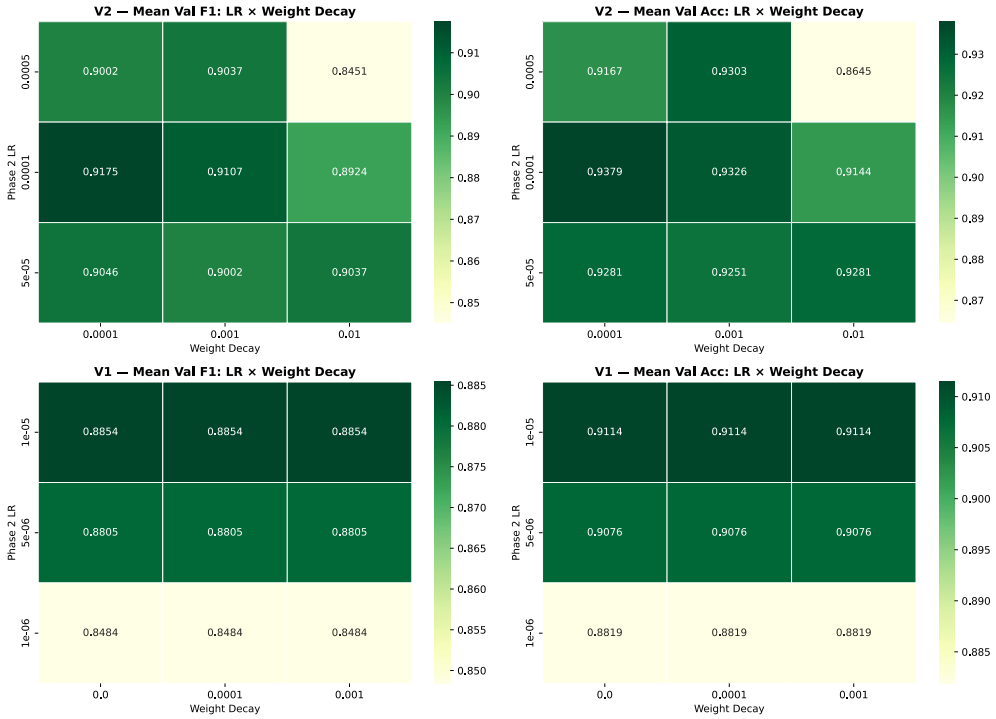


Figure 8: heatmap of ResNet18 phase 2 grid searches

- Hyperparameters selected via 3-fold grid search on the plain ResNet18 baseline
  - Only searched Phase 2 Learning rate and weight decay
  - Phase 1 parameters performed well in pilot testing and as such were kept
  - Grid search on baseline only – prevents architecture-specific tuning from confounding comparisons

- Head ablation: linear probe vs MLP head tested before any attention variants trained
  - Linear probe wins
  - Adopted for all future experiments, eliminating head capacity as a confound

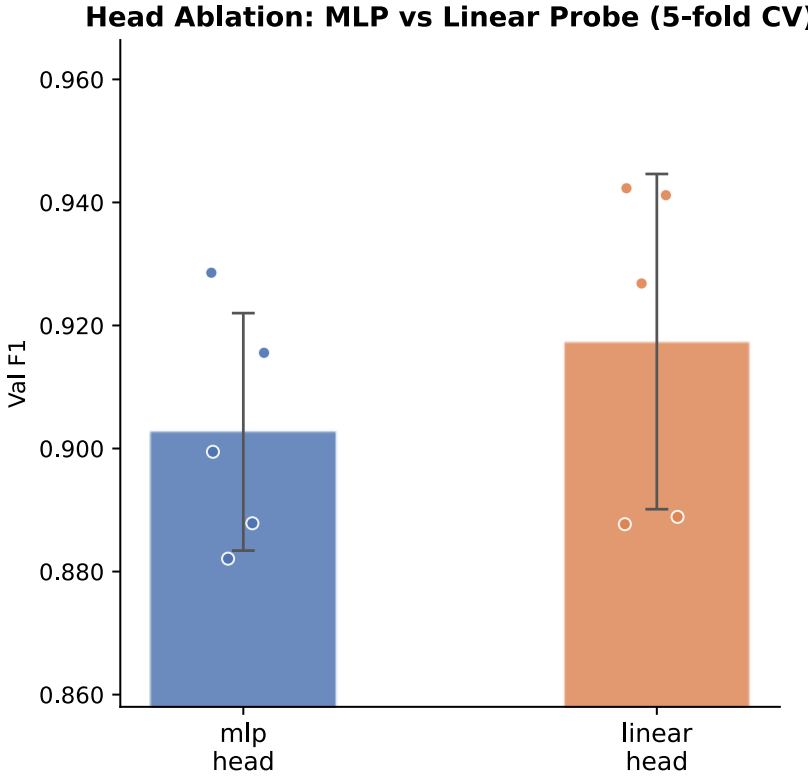
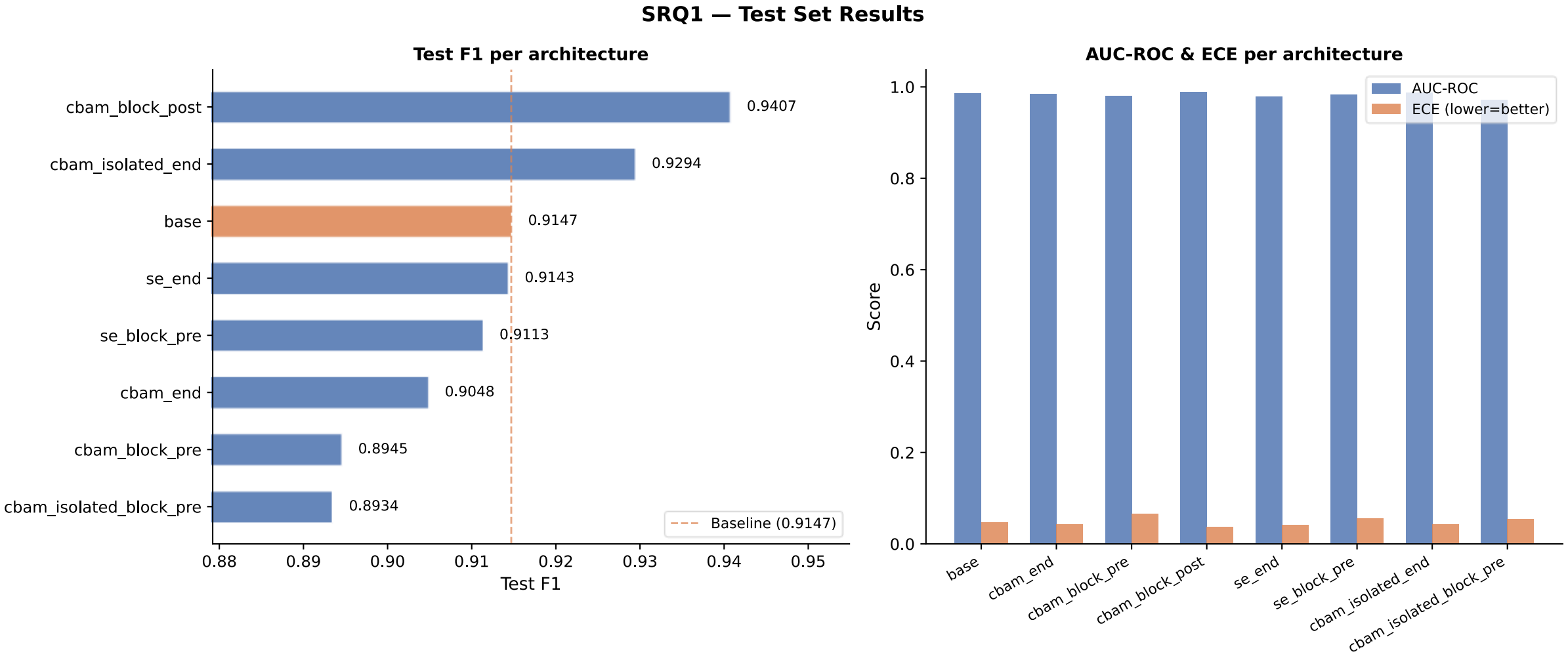
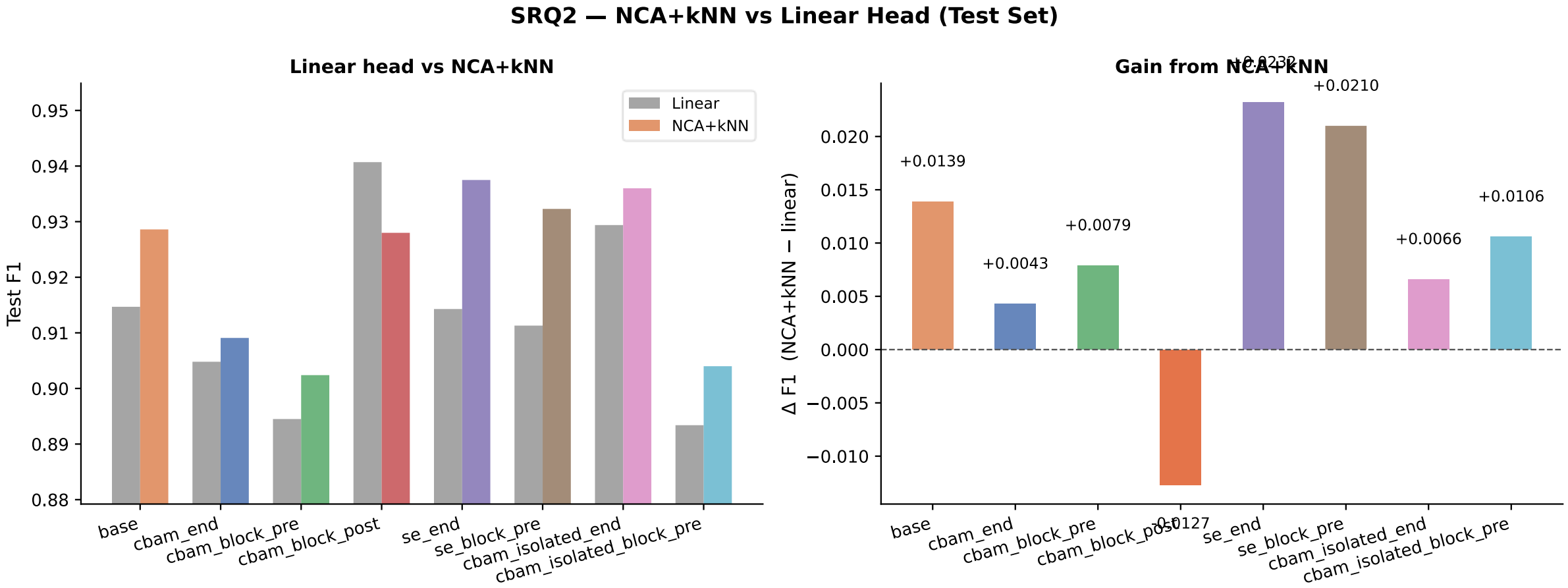


Figure 9: head ablation chart

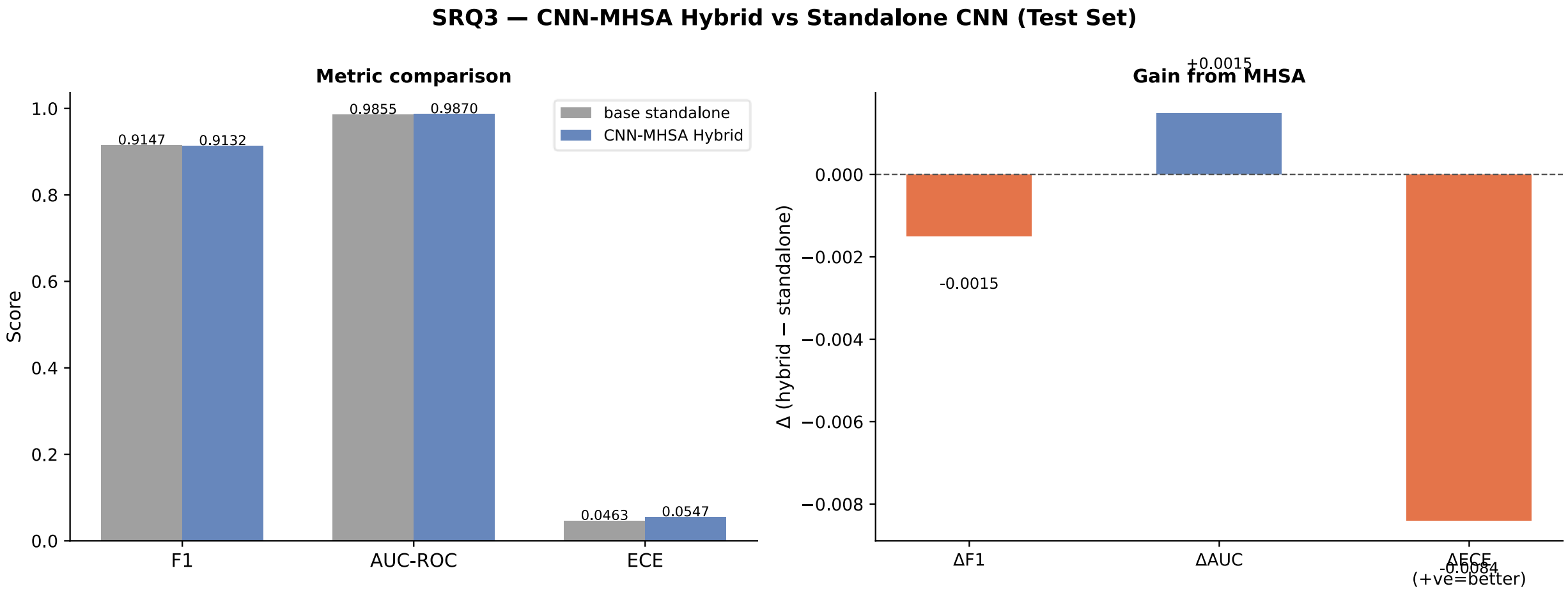
SRQ1 Attention Mechanisms



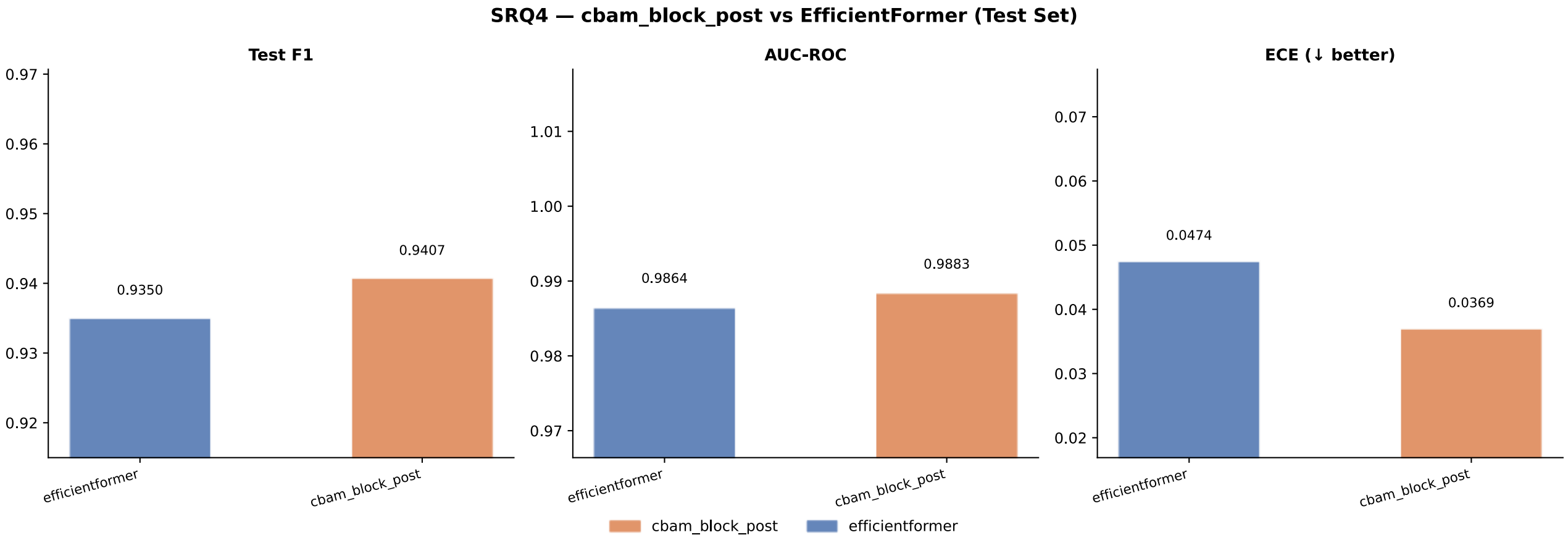
SRQ2 Metric Learning



SRQ3 Tokenised Self-Attention



SRQ4 ViT Comparison



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